

First Implementation of NLI Estimation on Coherent DSP Enabled by On-Chip Longitudinal Power Monitoring

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Abstract We report the first implementation of NLI estimation on a real-time coherent DSP, enabled by on-chip longitudinal power monitoring inside an OSFP. We demonstrate NLI estimation over a 450-km link using only standard 400ZR+ traffic, achieving a 0.62-dB maximum absolute error in generalized OSNR. ©2026 The Author(s)

Introduction

Coherent digital signal processors (DSPs) have progressively expanded their monitoring capabilities, evolving from cumulative link metrics to spatially-resolved diagnostics through longitudinal power monitoring (LPM) [1],[2]. Very recently, LPM has been realized fully on a coherent DSP chip inside an OSFP pluggable module [3], upgrading pluggable transceivers from pure transmission elements into distributed monitoring points without additional hardware.

Beyond the optical power profile itself, accurate estimation of nonlinear interference (NLI) is another crucial parameter for optical network operation and optimization, enabling noise-component segregation, launch power optimization, and margin design. Analytical NLI models such as the GN-model [4] have been widely adopted but require *a priori* knowledge of the link parameters. In contrast, estimation approaches driven by the received signal itself can naturally reflect the link state and thus potentially offer a more faithful estimate of the actual link NLI. Previous NLI estimation approaches have leveraged received constellations [5], frequency-domain analysis [6], or machine learning techniques [7]. More recently, the least-squares LPM has been applied to provide an accurate NLI estimation [8] as a by-product of the power profile computation [9] without requiring any special signal sequences. Subsequent studies have also addressed the impact of hard-decision and heterogeneous links [10], [11]. However, all of these NLI estimation demonstrations have relied on offline processing outside the transceiver, leaving the realization of LPM-based NLI estimation on a real coherent DSP chip unaddressed.

In this work, by extending the on-chip LPM in [3], we present the first implementation of LPM-based NLI estimation on a real-time coherent DSP chip inside a pluggable module. The NLI estimation arises as a natural extension of the on-chip LPM pipeline with negligible additional computation overhead. The on-chip NLI estimation is experimentally demonstrated over a 450-km link solely using standard 400ZR+ traffic, achieving a

maximum absolute error of 0.62 dB in generalized OSNR (GOSNR) prediction. These results add to coherent pluggable transceivers the NLI monitoring capability, allowing for in-transceiver SNR segregation and root cause analysis.

Implementation of NLI monitor on coherent DSP

In [9], an NLI estimation method based on the least-squares LPM algorithm [2] was proposed. In their method, the fibre-longitudinal power profile γ' is first computed from the received nonlinear signal A_1 and a perturbation matrix G describing nonlinear fibre propagation using the least squares algorithm [2]. The estimated power profile $\hat{\gamma}'$ is then substituted back to reconstruct nonlinear waveform as $\hat{A}_1 = G\hat{\gamma}'$ as a natural by-product of the least-squares LPM. From this, the nonlinear SNR associated with self-channel interference (SCI), SNR_{SCI} , is directly obtained as the ratio between the power of the transmitted reference signal and that of the estimated nonlinear term.

However, this approach requires storing the matrix G , whose memory footprint is prohibitive for DSP implementation as it ranges from megabytes to gigabytes. Consequently, the NLI estimation in [8]–[11] has been performed using offline processing outside the transceiver. To address this point, we implemented the LPM-based

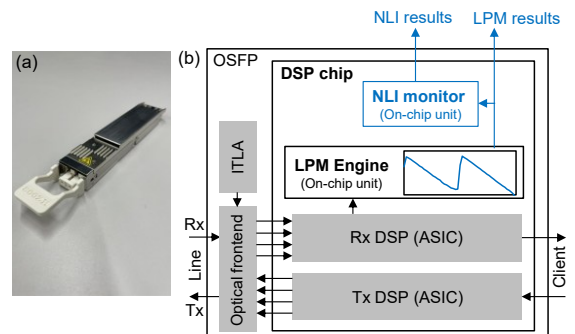


Fig. 1: (a) OSFP pluggable module with coherent DSP inside. (b) Block Diagram inside OSFP and DSP. The LPM and NLI monitor are implemented on DSP chip.

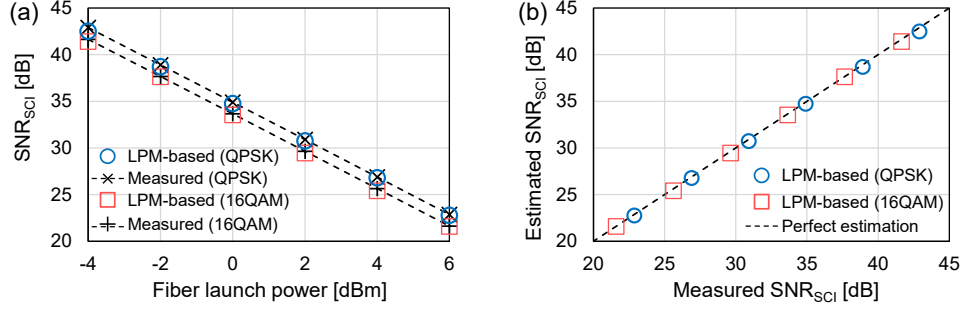


Fig. 2: Simulation results verifying the on-chip LPM-based NLI estimation algorithm for 3-span $\times$ 50-km link. (a) SNR_{SCI} versus fibre launch power for QPSK (400ZR+) and 16QAM (800ZR+). (b) Estimated versus measured SNR_{SCI} .

NLI estimation assisted by an analytical form inspired by [12]. This approach avoids storing the large matrix $\mathbf{G}$ during LPM and NLI computation while keeping the accuracy among various modulation formats, reducing the memory requirement by >99% and thereby bringing NLI estimation within the on-chip resource budget. In this way, the low memory LPM-based NLI monitor is integrated into the coherent DSP chip as shown in Fig. 1. The NLI estimation block reuses the intermediate quantities already computed inside the LPM engine, adding only a lightweight post-processing stage on the same DSP chip. Consequently, the memory footprint and power consumption of on-chip NLI estimation is negligible relative to that of LPM itself only. In addition, the transmitted signal required by LPM is reconstructed inside the Rx DSP from demodulated symbols, so both the LPM and NLI monitor can operate on live traffic without any dedicated training sequence.

To verify the implemented algorithm, we performed split-step simulations of the Manakov equation for a single-channel 118.2-GBd signal over 3 $\times$ 50-km spans ($\alpha = 0.2$ dB/km, $D = 17$ ps/nm/km, $\gamma = 1.3$ /W/km). Both QPSK (400ZR+) and 16QAM (800ZR+) were evaluated to confirm that the NLI monitor operates correctly across different modulation formats. The ASE noise corresponding to $\text{SNR}_{\text{ASE}} = 23$ dB was loaded at the receiver. Fig. 2(a) shows SNR_{SCI} as a function of fibre launch power. The "Measured" values were obtained from the received constellation without ASE loading, so that the constellation noise reflects only SCI. The "LPM-based" values were estimated by the on-chip NLI monitor operating under $\text{SNR}_{\text{ASE}} = 23$ dB. For both QPSK and 16QAM, the LPM-based estimates closely follow the measured SNR_{SCI} across the entire launch-power range, confirming that the NLI monitor correctly extracts SCI even in the presence of ASE noise and properly decouples SCI and ASE contributions. Fig. 2(b) plots the estimated versus measured SNR_{SCI} , showing agreement over a ~ 20 -dB dynamic range with a maximum absolute

error of 0.40 dB.

Experimental setup

To experimentally validate the NLI monitor, we conducted a single-channel transmission experiment (Fig. 3(a)). For the transmitter, a coherent pluggable OSFP module is used, transmitting a standard 400ZR+ dual-polarization QPSK signal at 118.2 GBd. The channel under test (CUT) is set at a centre frequency of 193.7 THz. The transmission link consists of 6 spans of 75-km G.652.D fibres (average attenuation 0.186 dB/km, chromatic dispersion 16.80 ps/nm/km), making a total link length of 450 km. The span losses are fully compensated by EDFAs, and the fibre launch power is swept in 2-dB steps around the Q-factor maximizing point (Fig. 3(b)). After transmission, the CUT is received by the same OSFP pluggable module, which performs signal demodulation, on-chip LPM, and on-chip NLI estimation concurrently within the Rx DSP chip. The transmitted signal necessary for LPM is reconstructed within the Rx DSP chip and is not assumed to be known a priori; no special training sequence is used. The estimated SNR_{SCI} values, together with the power profile, are read out through the standard I²C interface of the module.

Experimental Results

Fig. 4(a) shows the on-chip LPM results at fibre launch powers swept in 2-dB steps. The characteristic sawtooth pattern of span-by-span EDFA amplification and fibre loss is clearly resolved over all six spans, demonstrating that the on-chip LPM engine reliably captures the power profile across a wide launch-power range using in-service signals. Fig. 4(b) plots the corresponding SNR components as a function of average fibre launch power. Unlike the simulation in Fig. 2, directly measuring SNR_{SCI} as an experimental reference is impractical as it is largely affected by the dominant transceiver SNR, particularly in the low launch power and weak SCI regime. We therefore validate the NLI monitor at the GOSNR level, comparing the proposed GOSNR ($\text{SNR}_{\text{SCI}} + \text{SNR}_{\text{ASE}}$) with the reference GOSNR ($\text{SNR}_{\text{tot}} - \text{SNR}_{\text{B2B}}$), where SNR_{ASE} is obtained from an

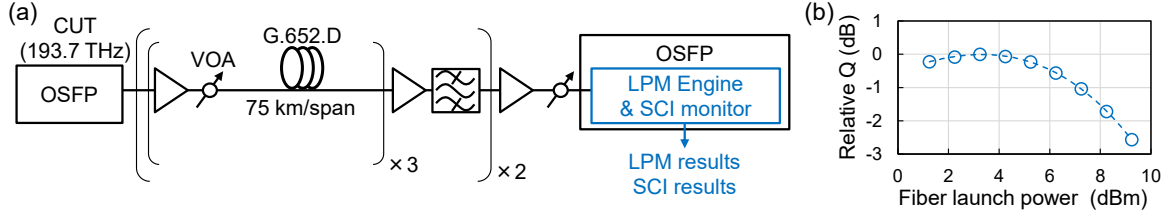


Fig. 3: (a) Experimental setup for 6-span $\times$ 75-km ($=$ 450 km) transmission. (b) Relative Q value vs fibre launch power after 450-km transmission.

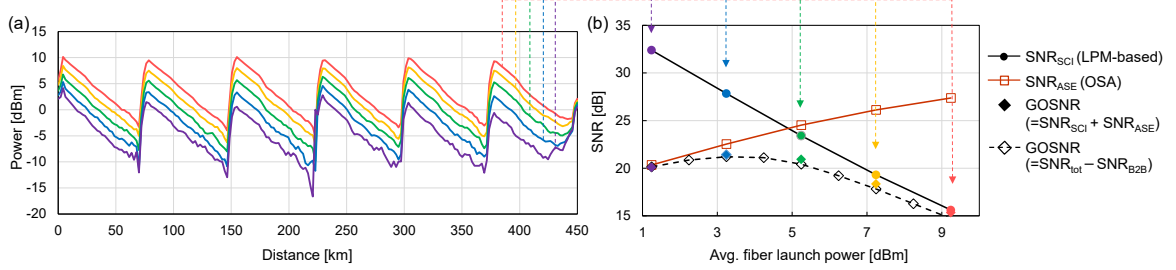


Fig. 4: Experimental results over 6-span $\times$ 75-km link. (a) On-chip LPM results at fibre launch powers swept in 2-dB steps; colours correspond to the filled markers in (b). (b) SNR_{SCI} from the proposed NLI monitor, SNR_{ASE} , and GOSNR obtained as $\text{SNR}_{\text{SCI}} + \text{SNR}_{\text{ASE}}$ (proposed) and as $\text{SNR}_{\text{tot}} - \text{SNR}_{\text{B2B}}$ (reference), plotted against average fibre launch power. The proposed and reference GOSNR values are in good agreement with a maximum absolute error of 0.62 dB.

optical spectrum analyser, SNR_{tot} is the value reported by the OSFP transceiver, and SNR_{B2B} is measured in advance in a back-to-back configuration. SNR_{SCI} , estimated by the on-chip NLI monitor from the LPM profiles in (a), decreases with launch power at the expected slope of ~ -2 dB/dB, while SNR_{ASE} grows as the signal power increases. The resulting proposed GOSNR agrees well with the reference across the wide launch-power range with a maximum absolute error of 0.62 dB, confirming that the on-chip NLI monitor correctly estimates the nonlinear noise. Importantly, this is achieved entirely within the transceiver DSP, without any external instrumentation or offline processing, demonstrating the feasibility of NLI monitoring in deployed pluggable modules.

Discussion and Applications

The present implementation estimates only the SCI component of NLI. Cross-channel interference (XCI) from neighbouring WDM channels is not directly captured, since on-chip LPM provides the longitudinal power profile of the self channel alone, which is a fundamental constraint of single-transceiver monitoring. Despite this restriction, the on-chip SCI monitor enables several practical applications.

First, the on-chip NLI monitor enables SNR segregation inside the transceiver. The total SNR reported by conventional DSPs aggregates linear and nonlinear impairments, making it difficult to attribute margin changes to a specific noise source. By decomposing the total SNR into SNR_{SCI} and others directly on-chip, operators can identify whether an observed SNR degradation

originates from excessive or insufficient self-channel launch power and how much margin is available just by accessing transceivers, enabling more precise margin management than total-SNR-based monitoring alone.

Second, although only the cumulative SCI is presented in this work, the underlying LPM carries distance-resolved information and therefore allows per-span SCI estimation as demonstrated in [13]. This enables distributed nonlinear performance telemetry in which each pluggable transceiver identifies the specific span that contributes most to the accumulated SCI, supporting root-cause analysis of SNR degradations and targeted maintenance in deployed networks.

Conclusions

We have presented the first implementation of LPM-based NLI monitor that estimates self-channel nonlinear interference from spatial power profiles estimated by on-chip LPM. This NLI monitor is fully implemented within a coherent DSP inside a pluggable optics and is experimentally demonstrated over a 450-km link using standard ZR+ traffic, achieving a maximum absolute error of 0.62 dB in GOSNR. These results establish LPM-based on-chip NLI estimation as a new practical link monitor available in DSP and coherent pluggable optics whose application reach ranges from DCI to long haul, thereby enhancing physical-layer telemetry for operational tasks such as margin management and root cause analysis in a wide range of optical network infrastructure.

Acknowledgements

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